

Spring Framework & Microservices Interview Questions - Comprehensive List

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1. Spring Framework Core

Fundamentals

1. What is the Spring Framework? Why is it so popular?
2. What are the main features and benefits of Spring Framework?
3. What are the different modules in Spring Framework?
4. What is Inversion of Control (IoC)?
5. What is Dependency Injection (DI)?
6. What are the types of Dependency Injection in Spring?
7. What is the difference between IoC and DI?
8. Explain the Spring IoC container.
9. What is the difference between BeanFactory and ApplicationContext?
10. Which one should you use: BeanFactory or ApplicationContext?

Spring Beans

11. What is a Spring Bean?
12. What is the lifecycle of a Spring Bean?
13. What are the different bean scopes in Spring?
14. What is the default bean scope in Spring?
15. Explain the singleton and prototype bean scopes.
16. What is the difference between singleton and prototype scope?

17. How do you create a Spring Bean?
18. What is bean wiring in Spring?
19. What is autowiring in Spring?
20. What are the different autowiring modes in Spring?
21. What is the difference between @Autowired and @Inject?
22. What is the difference between @Autowired and @Resource?
23. What is @Qualifier annotation used for?
24. What is @Primary annotation?
25. What is constructor injection and setter injection?
26. Which is better: constructor injection or setter injection?

Spring Configuration

27. What is a Spring configuration file?
28. What are the different ways to configure Spring application?
29. What is XML-based configuration?
30. What is Java-based configuration?
31. What is annotation-based configuration?
32. What is @Configuration annotation?
33. What is @Bean annotation?
34. What is @Component annotation?
35. What is @ComponentScan annotation?
36. What is the difference between @Component, @Repository, @Service, and @Controller?
37. What is the difference between @Component and @Bean?
38. What is @Value annotation?
39. What is @PropertySource annotation?
40. How do you externalize configuration in Spring?

Advanced Core Concepts

41. What is lazy initialization in Spring?
 42. What is @Lazy annotation?
 43. What are inner beans in Spring?
 44. What is a BeanPostProcessor?
 45. What is a BeanFactoryPostProcessor?
 46. What is the difference between BeanPostProcessor and BeanFactoryPostProcessor?
 47. What is ApplicationContext event handling?
 48. What are the different types of events in Spring?
 49. What is @EventListener annotation?
 50. What is the difference between Spring and Java EE?
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2. Spring Boot

Spring Boot Fundamentals

51. What is Spring Boot?
52. What are the advantages of Spring Boot over Spring Framework?
53. What is the difference between Spring and Spring Boot?
54. What problem does Spring Boot solve?
55. What are the main features of Spring Boot?
56. What is auto-configuration in Spring Boot?
57. How does Spring Boot auto-configuration work?
58. What is @SpringBootApplication annotation?
59. What does @EnableAutoConfiguration do?
60. What is the difference between @SpringBootApplication and @EnableAutoConfiguration?

Spring Boot Starters

61. What are Spring Boot starters?
62. What is spring-boot-starter-parent?
63. What are the most commonly used Spring Boot starters?
64. What is spring-boot-starter-web?
65. What is spring-boot-starter-data-jpa?
66. What is spring-boot-starter-test?
67. What is spring-boot-starter-security?
68. How do Spring Boot starters help in development?
69. Can you create your own custom starter?

Spring Boot Configuration

70. What is application.properties file?
71. What is application.yml file?
72. What is the difference between application.properties and application.yml?
73. How do you change the default port in Spring Boot?
74. What are Spring Boot profiles?
75. How do you define different profiles in Spring Boot?
76. How do you activate a specific profile?
77. What is @ConfigurationProperties annotation?
78. What is @Value annotation and how is it used?
79. How do you override default configuration in Spring Boot?
80. What is spring.factories file?

Embedded Servers

81. What are embedded servers in Spring Boot?
82. What is the default embedded server in Spring Boot?

- 83. What are the different embedded servers supported by Spring Boot?
- 84. How do you change the embedded server from Tomcat to Jetty?
- 85. How do you disable the embedded server?
- 86. Can you deploy a Spring Boot application on an external server?
- 87. What is the difference between JAR and WAR packaging?

Spring Boot DevTools

- 88. What is Spring Boot DevTools?
- 89. What are the features of Spring Boot DevTools?
- 90. What is LiveReload in DevTools?
- 91. What is automatic restart in DevTools?
- 92. How does DevTools improve developer productivity?

Spring Boot Advanced

- 93. What is @Conditional annotation?
 - 94. What are the different @Conditional annotations in Spring Boot?
 - 95. What is @ConditionalOnClass?
 - 96. What is @ConditionalOnMissingBean?
 - 97. What is @ConditionalOnProperty?
 - 98. How do you create a custom auto-configuration?
 - 99. What is the Spring Boot Banner?
 - 100. How do you customize the Spring Boot Banner?
 - 101. What is CommandLineRunner interface?
 - 102. What is ApplicationRunner interface?
 - 103. What is the difference between CommandLineRunner and ApplicationRunner?
 - 104. How do you execute code at application startup?
 - 105. What is @PostConstruct annotation?
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3. Spring MVC

Spring MVC Fundamentals

- 106. What is Spring MVC?
- 107. What is MVC design pattern?
- 108. What are the main components of Spring MVC?
- 109. What is DispatcherServlet?
- 110. What is the role of DispatcherServlet in Spring MVC?
- 111. Explain the request flow in Spring MVC.
- 112. What is HandlerMapping?
- 113. What is HandlerAdapter?
- 114. What is ViewResolver?
- 115. What is the difference between @Controller and @RestController?

Request Mapping

- 116. What is @RequestMapping annotation?
- 117. What are the attributes of @RequestMapping?
- 118. What is @GetMapping?
- 119. What is @PostMapping?
- 120. What is @PutMapping?
- 121. What is @DeleteMapping?
- 122. What is @PatchMapping?
- 123. What is the difference between @RequestMapping and @GetMapping?
- 124. How do you handle path variables in Spring MVC?
- 125. What is @PathVariable annotation?
- 126. What is @RequestParam annotation?
- 127. What is the difference between @PathVariable and @RequestParam?
- 128. What is @RequestBody annotation?
- 129. What is @ResponseBody annotation?
- 130. What is @RequestHeader annotation?

Request & Response Handling

- 131. How do you return JSON from a Spring MVC controller?
- 132. How do you consume JSON in Spring MVC?
- 133. What is HttpMessageConverter?
- 134. What is the role of Jackson in Spring Boot?
- 135. How do you handle form data in Spring MVC?
- 136. What is @ModelAttribute annotation?
- 137. What is Model in Spring MVC?
- 138. What is ModelAndView?
- 139. What is the difference between Model and ModelAndView?
- 140. How do you handle file uploads in Spring MVC?
- 141. What is MultipartFile?

Exception Handling

- 142. How do you handle exceptions in Spring MVC?
- 143. What is @ExceptionHandler annotation?
- 144. What is @ControllerAdvice?
- 145. What is @RestControllerAdvice?
- 146. What is the difference between @ControllerAdvice and @RestControllerAdvice?
- 147. How do you create a global exception handler?
- 148. What is ResponseEntityExceptionHandler?
- 149. How do you return custom error responses?
- 150. What is @ResponseStatus annotation?

Validation

- 151. How do you validate request data in Spring MVC?
- 152. What is @Valid annotation?
- 153. What is @Validated annotation?
- 154. What is the difference between @Valid and @Validated?
- 155. What are the common validation annotations?
- 156. What is BindingResult?
- 157. How do you create custom validation annotations?
- 158. What is ConstraintValidator?

Interceptors & Filters

- 159. What is an interceptor in Spring MVC?
 - 160. What is HandlerInterceptor?
 - 161. What are the methods in HandlerInterceptor?
 - 162. What is the difference between Filter and Interceptor?
 - 163. How do you register an interceptor?
 - 164. When should you use Filter vs Interceptor?
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4. Spring Data JPA

JPA Fundamentals

- 165. What is JPA?
- 166. What is Hibernate?
- 167. What is the difference between JPA and Hibernate?
- 168. What is Spring Data JPA?
- 169. What are the advantages of Spring Data JPA?
- 170. What is an Entity in JPA?
- 171. What is @Entity annotation?
- 172. What is @Table annotation?
- 173. What is @Id annotation?
- 174. What is @GeneratedValue annotation?
- 175. What are the different GenerationType strategies?

Repository Pattern

- 176. What is a Repository in Spring Data JPA?
- 177. What is the Repository interface?
- 178. What is CrudRepository?
- 179. What is JpaRepository?
- 180. What is the difference between CrudRepository and JpaRepository?
- 181. What is PagingAndSortingRepository?
- 182. Which repository should you extend and when?
- 183. How does Spring Data JPA implement repository interfaces?

184. What is @Repository annotation?

Query Methods

- 185. What are query methods in Spring Data JPA?
- 186. How do you create custom finder methods?
- 187. What are the naming conventions for query methods?
- 188. What is @Query annotation?
- 189. How do you write custom queries using @Query?
- 190. What is JPQL?
- 191. What is the difference between JPQL and SQL?
- 192. How do you write native queries in Spring Data JPA?
- 193. What is @NamedQuery?
- 194. What is @Param annotation in queries?
- 195. How do you implement pagination in Spring Data JPA?
- 196. How do you implement sorting in Spring Data JPA?
- 197. What is Pageable interface?
- 198. What is Page and Slice?
- 199. What is the difference between Page and Slice?

Entity Relationships

- 200. What are the different types of relationships in JPA?
- 201. What is @OneToOne relationship?
- 202. What is @OneToMany relationship?
- 203. What is @ManyToOne relationship?
- 204. What is @ManyToMany relationship?
- 205. What is @JoinColumn annotation?
- 206. What is @JoinTable annotation?
- 207. What is mappedBy attribute?
- 208. What is the difference between unidirectional and bidirectional relationships?
- 209. What is CascadeType?
- 210. What are the different cascade types?
- 211. What is FetchType?
- 212. What is the difference between EAGER and LAZY loading?
- 213. What is the N+1 select problem?
- 214. How do you solve the N+1 select problem?

Advanced JPA

- 215. What is @Transactional annotation in Spring Data JPA?
- 216. What is EntityManager?
- 217. What is the difference between save() and saveAndFlush()?
- 218. What is @Modifying annotation?
- 219. How do you perform bulk operations?
- 220. What is @EntityGraph?

- 221. What is Criteria API?
 - 222. What is Specification interface?
 - 223. How do you implement dynamic queries?
 - 224. What is @Embedded annotation?
 - 225. What is @Embeddable annotation?
 - 226. What is composite primary key?
 - 227. What is @EmbeddedId and @IdClass?
 - 228. What is @Version annotation and optimistic locking?
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5. Spring Security

Security Fundamentals

- 229. What is Spring Security?
- 230. What are the main features of Spring Security?
- 231. What is authentication?
- 232. What is authorization?
- 233. What is the difference between authentication and authorization?
- 234. What is a SecurityContext?
- 235. What is SecurityContextHolder?
- 236. What is Authentication object?
- 237. What is Principal?
- 238. What is GrantedAuthority?

Configuration

- 239. How do you enable Spring Security in a Spring Boot application?
- 240. What is WebSecurityConfigurerAdapter?
- 241. What is the configure method in Spring Security?
- 242. How do you configure HTTP security?
- 243. What is @EnableWebSecurity annotation?
- 244. What is @EnableGlobalMethodSecurity?
- 245. What are the different types of security expressions?
- 246. What is hasRole() and hasAuthority()?
- 247. What is the difference between ROLE_ prefix and authority?

Authentication

- 248. What are the different authentication mechanisms in Spring Security?
- 249. What is form-based authentication?
- 250. What is HTTP Basic authentication?
- 251. What is in-memory authentication?
- 252. What is JDBC authentication?
- 253. What is UserDetailsService?
- 254. What is UserDetails interface?

- 255. How do you implement custom authentication?
- 256. What is AuthenticationManager?
- 257. What is AuthenticationProvider?
- 258. What is PasswordEncoder?
- 259. What are the different PasswordEncoder implementations?
- 260. What is BCryptPasswordEncoder?
- 261. Why should passwords be encoded?
- 262. What is password hashing and salting?

Authorization

- 263. How do you configure method-level security?
- 264. What is @PreAuthorize annotation?
- 265. What is @PostAuthorize annotation?
- 266. What is @Secured annotation?
- 267. What is the difference between @Secured and @PreAuthorize?
- 268. What is @RolesAllowed annotation?
- 269. How do you implement role-based access control (RBAC)?
- 270. How do you implement custom authorization logic?

Security Filters

- 271. What is the Spring Security filter chain?
- 272. What is DelegatingFilterProxy?
- 273. What are the important filters in Spring Security?
- 274. What is UsernamePasswordAuthenticationFilter?
- 275. What is BasicAuthenticationFilter?
- 276. What is the order of filters in the security chain?
- 277. How do you add custom filters?
- 278. What is OncePerRequestFilter?

JWT & OAuth2

- 279. What is JWT (JSON Web Token)?
- 280. What are the components of JWT?
- 281. How do you implement JWT authentication in Spring Boot?
- 282. What is OAuth 2.0?
- 283. What are the different OAuth 2.0 grant types?
- 284. What is authorization code grant?
- 285. What is client credentials grant?
- 286. What is Resource Server?
- 287. What is Authorization Server?
- 288. How do you implement OAuth 2.0 in Spring Security?
- 289. What is Spring Security OAuth2?
- 290. What is @EnableResourceServer?
- 291. What is @EnableAuthorizationServer?

CORS & CSRF

- 292. What is CORS?
- 293. How do you configure CORS in Spring Security?
- 294. What is CSRF?
- 295. How does Spring Security protect against CSRF?
- 296. When should you disable CSRF protection?
- 297. What is @CrossOrigin annotation?

Advanced Security

- 298. What is remember-me authentication?
 - 299. What is session fixation protection?
 - 300. What is session management in Spring Security?
 - 301. How do you implement logout functionality?
 - 302. What is SecurityContextPersistenceFilter?
 - 303. How do you test Spring Security?
 - 304. What is @WithMockUser?
 - 305. What is @WithUserDetails?
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6. Spring Cloud

Spring Cloud Fundamentals

- 306. What is Spring Cloud?
- 307. What problems does Spring Cloud solve?
- 308. What are the main components of Spring Cloud?
- 309. What is the difference between Spring Boot and Spring Cloud?
- 310. What is a cloud-native application?
- 311. What is the 12-factor app methodology?

Service Discovery

- 312. What is service discovery?
- 313. Why is service discovery important in microservices?
- 314. What is Eureka?
- 315. What is Eureka Server?
- 316. What is Eureka Client?
- 317. What is @EnableEurekaServer?
- 318. What is @EnableDiscoveryClient?
- 319. How does Eureka work?
- 320. What is service registration?
- 321. What is the heartbeat mechanism in Eureka?
- 322. What is self-preservation mode in Eureka?
- 323. What are alternatives to Eureka?
- 324. What is Consul?

325. What is Zookeeper for service discovery?

API Gateway

- 326. What is an API Gateway?
- 327. Why do you need an API Gateway in microservices?
- 328. What is Spring Cloud Gateway?
- 329. What is Zuul?
- 330. What is the difference between Zuul and Spring Cloud Gateway?
- 331. Which is better: Zuul or Spring Cloud Gateway?
- 332. What are routes in Spring Cloud Gateway?
- 333. What are predicates in Spring Cloud Gateway?
- 334. What are filters in Spring Cloud Gateway?
- 335. What is the difference between pre-filter and post-filter?
- 336. How do you configure routes in Spring Cloud Gateway?
- 337. What is path rewriting in API Gateway?
- 338. How do you implement authentication in API Gateway?
- 339. How do you implement rate limiting in API Gateway?

Configuration Management

- 340. What is centralized configuration?
- 341. Why is external configuration important?
- 342. What is Spring Cloud Config?
- 343. What is Config Server?
- 344. What is Config Client?
- 345. How does Spring Cloud Config work?
- 346. What is @EnableConfigServer?
- 347. Where can you store configuration files?
- 348. How do you use Git as a configuration repository?
- 349. What is @RefreshScope?
- 350. How do you refresh configuration without restarting?
- 351. What is Spring Cloud Bus?
- 352. How does Spring Cloud Bus help in configuration refresh?
- 353. What is /actuator/refresh endpoint?
- 354. What is encryption and decryption in Config Server?

Load Balancing

- 355. What is load balancing?
- 356. What is client-side load balancing?
- 357. What is server-side load balancing?
- 358. What is Ribbon?
- 359. What is Spring Cloud LoadBalancer?
- 360. What is the difference between Ribbon and Spring Cloud LoadBalancer?
- 361. What is @LoadBalanced annotation?
- 362. How does @LoadBalanced work with RestTemplate?

- 363. What are load balancing algorithms?
- 364. What is round-robin load balancing?

Circuit Breaker

- 365. What is a circuit breaker pattern?
- 366. Why do you need circuit breakers in microservices?
- 367. What is Hystrix?
- 368. What is Resilience4j?
- 369. What is the difference between Hystrix and Resilience4j?
- 370. Which is recommended: Hystrix or Resilience4j?
- 371. What are the three states of a circuit breaker?
- 372. What is @CircuitBreaker annotation?
- 373. What is a fallback method?
- 374. How do you implement circuit breaker with Resilience4j?
- 375. What is bulkhead pattern?
- 376. What is rate limiter?
- 377. What is retry mechanism?
- 378. What is timeout configuration?

Distributed Tracing

- 379. What is distributed tracing?
- 380. Why is distributed tracing important?
- 381. What is Spring Cloud Sleuth?
- 382. What is Zipkin?
- 383. What is Jaeger?
- 384. What is a trace ID?
- 385. What is a span ID?
- 386. How does Spring Cloud Sleuth work?
- 387. How do you integrate Sleuth with Zipkin?
- 388. What is OpenTelemetry?
- 389. What is the difference between Sleuth and OpenTelemetry?

Messaging

- 390. What is Spring Cloud Stream?
- 391. What is a message broker?
- 392. What is the difference between Kafka and RabbitMQ?
- 393. What is a binder in Spring Cloud Stream?
- 394. What is @EnableBinding?
- 395. What is @StreamListener?
- 396. What are channels in Spring Cloud Stream?
- 397. What is event-driven architecture?

7. Microservices Architecture

Microservices Fundamentals

- 398. What is microservices architecture?
- 399. What is the difference between monolithic and microservices architecture?
- 400. What are the advantages of microservices?
- 401. What are the disadvantages of microservices?
- 402. When should you use microservices?
- 403. When should you NOT use microservices?
- 404. What are the characteristics of microservices?
- 405. What is domain-driven design (DDD)?
- 406. What is bounded context?
- 407. How do you identify microservices boundaries?
- 408. What is single responsibility principle in microservices?

Microservices Design

- 409. What is the database per service pattern?
- 410. What are shared databases in microservices?
- 411. What is polyglot persistence?
- 412. What is eventual consistency?
- 413. What is the CAP theorem?
- 414. How do you maintain data consistency across microservices?
- 415. What is a saga pattern?
- 416. What are the types of saga patterns?
- 417. What is choreography-based saga?
- 418. What is orchestration-based saga?
- 419. What is the difference between choreography and orchestration?
- 420. What is CQRS (Command Query Responsibility Segregation)?
- 421. What is event sourcing?
- 422. What is the difference between CQRS and event sourcing?

Microservices Communication

- 423. What are the different ways microservices can communicate?
- 424. What is synchronous communication?
- 425. What is asynchronous communication?
- 426. What is the difference between synchronous and asynchronous communication?
- 427. When should you use synchronous vs asynchronous communication?
- 428. What is REST in microservices?
- 429. What is gRPC?
- 430. What is the difference between REST and gRPC?
- 431. What is GraphQL?
- 432. What is message-driven communication?
- 433. What is publish-subscribe pattern?

434. What is request-response pattern?

Service Mesh

- 435. What is a service mesh?
- 436. Why do you need a service mesh?
- 437. What is Istio?
- 438. What is Linkerd?
- 439. What are the features of a service mesh?
- 440. What is a sidecar proxy?
- 441. What is Envoy proxy?
- 442. What is traffic management in service mesh?
- 443. What is mutual TLS (mTLS)?
- 444. What is observability in service mesh?

Microservices Testing

- 445. What are the different levels of testing in microservices?
- 446. What is contract testing?
- 447. What is consumer-driven contract testing?
- 448. What is Spring Cloud Contract?
- 449. What is Pact?
- 450. What is integration testing in microservices?
- 451. What is component testing?
- 452. What is end-to-end testing?
- 453. How do you test microservices in isolation?
- 454. What is test pyramid in microservices?
- 455. What is chaos engineering?

Microservices Challenges

- 456. What are the challenges of microservices architecture?
- 457. What is network latency in microservices?
- 458. What is cascading failure?
- 459. How do you handle distributed transactions?
- 460. What is the two-phase commit (2PC) protocol?
- 461. Why is 2PC not recommended in microservices?
- 462. What is distributed logging?
- 463. How do you aggregate logs from multiple microservices?
- 464. What is the ELK stack?
- 465. What is correlation ID?
- 466. How do you debug issues in microservices?

8. Microservices Design Patterns

Decomposition Patterns

- 467. What is decomposition by business capability?
- 468. What is decomposition by subdomain?
- 469. What is the Strangler Fig pattern?
- 470. When should you use the Strangler pattern?

Integration Patterns

- 471. What is the API Gateway pattern?
- 472. What is the Backend for Frontend (BFF) pattern?
- 473. What is the Aggregator pattern?
- 474. What is the Proxy pattern in microservices?
- 475. What is the Chained pattern?

Database Patterns

- 476. What is the Database per Service pattern?
- 477. What is the Shared Database pattern?
- 478. What is the Saga pattern for distributed transactions?
- 479. What is the Event Sourcing pattern?
- 480. What is the CQRS pattern?
- 481. What is the Outbox pattern?

Observability Patterns

- 482. What is the Log Aggregation pattern?
- 483. What is the Distributed Tracing pattern?
- 484. What is the Health Check pattern?
- 485. What is the Audit Logging pattern?
- 486. What is the Exception Tracking pattern?

Cross-Cutting Concerns Patterns

- 487. What is the Externalized Configuration pattern?
- 488. What is the Service Discovery pattern?
- 489. What is the Circuit Breaker pattern?
- 490. What is the Bulkhead pattern?
- 491. What is the Rate Limiting pattern?
- 492. What is the Retry pattern?
- 493. What is the Timeout pattern?

Deployment Patterns

- 494. What is the Multiple Service Instances per Host pattern?
- 495. What is the Service Instance per Host pattern?

- 496. What is the Service Instance per Container pattern?
- 497. What is the Serverless Deployment pattern?
- 498. What is Blue-Green Deployment?
- 499. What is Canary Deployment?
- 500. What is Rolling Deployment?

Security Patterns

- 501. What is the Access Token pattern?
- 502. What is the API Key pattern?
- 503. What is the Service-to-Service authentication pattern?

UI Patterns

- 504. What is Server-Side Page Fragment Composition?
- 505. What is Client-Side UI Composition?
- 506. What is Micro Frontends?

9. RESTful Web Services

REST Fundamentals

- 507. What is REST?
- 508. What are RESTful web services?
- 509. What are the principles of REST?
- 510. What are the characteristics of RESTful services?
- 511. What is a resource in REST?
- 512. What is a URI?
- 513. What is the difference between URI and URL?
- 514. What are HTTP methods?
- 515. What is the purpose of GET, POST, PUT, DELETE, PATCH methods?
- 516. What is idempotency?
- 517. Which HTTP methods are idempotent?
- 518. What is the difference between PUT and PATCH?
- 519. What is the difference between PUT and POST?

HTTP Status Codes

- 520. What are HTTP status codes?
- 521. What are the different categories of HTTP status codes?
- 522. What is the difference between 200 and 201 status codes?
- 523. What is 204 No Content?
- 524. What is 400 Bad Request?
- 525. What is 401 Unauthorized?
- 526. What is 403 Forbidden?
- 527. What is the difference between 401 and 403?

- 528. What is 404 Not Found?
- 529. What is 500 Internal Server Error?
- 530. What is 503 Service Unavailable?

REST API Design

- 531. What are REST API best practices?
- 532. How do you version REST APIs?
- 533. What are the different ways to version APIs?
- 534. What is HATEOAS?
- 535. What is Richardson Maturity Model?
- 536. What is content negotiation?
- 537. What is the Accept header?
- 538. What is the Content-Type header?
- 539. How do you handle pagination in REST APIs?
- 540. How do you handle filtering and sorting in REST APIs?
- 541. What is rate limiting in REST APIs?
- 542. How do you document REST APIs?
- 543. What is Swagger/OpenAPI?
- 544. What is @ApiOperation annotation?

REST Security

- 545. How do you secure REST APIs?
- 546. What is API authentication?
- 547. What is API authorization?
- 548. What is token-based authentication?
- 549. What is session-based authentication?
- 550. What is the difference between stateless and stateful authentication?
- 551. Why are REST APIs stateless?
- 552. What is Bearer token?
- 553. How do you implement API key authentication?

REST Client

- 554. What is RestTemplate?
- 555. What is WebClient?
- 556. What is the difference between RestTemplate and WebClient?
- 557. Which is better: RestTemplate or WebClient?
- 558. What is Feign Client?
- 559. What is @FeignClient annotation?
- 560. How does Feign work with Eureka?
- 561. What is the advantage of using Feign?
- 562. How do you handle errors in REST clients?

10. Spring Boot Actuator & Monitoring

Actuator Fundamentals

- 563. What is Spring Boot Actuator?
- 564. What are the features of Spring Boot Actuator?
- 565. How do you enable Actuator in Spring Boot?
- 566. What are actuator endpoints?
- 567. What are the most commonly used actuator endpoints?
- 568. What is /actuator/health endpoint?
- 569. What is /actuator/info endpoint?
- 570. What is /actuator/metrics endpoint?
- 571. What is /actuator/env endpoint?
- 572. What is /actuator/loggers endpoint?
- 573. What is /actuator/beans endpoint?
- 574. What is /actuator/mappings endpoint?
- 575. How do you enable/disable actuator endpoints?
- 576. How do you secure actuator endpoints?
- 577. How do you customize actuator endpoints?

Health Checks

- 578. What is a health indicator?
- 579. What are the different health indicator states?
- 580. How do you create a custom health indicator?
- 581. What is @Component annotation for health indicators?
- 582. What is liveness and readiness probes?
- 583. How do you configure health groups?

Metrics & Monitoring

- 584. What is Micrometer?
- 585. What are the different types of metrics?
- 586. How do you create custom metrics?
- 587. What is a counter metric?
- 588. What is a gauge metric?
- 589. What is a timer metric?
- 590. How do you integrate Actuator with Prometheus?
- 591. How do you integrate Actuator with Grafana?
- 592. What is the /actuator/prometheus endpoint?

Application Monitoring

- 593. What is APM (Application Performance Monitoring)?
- 594. What is New Relic?
- 595. What is Dynatrace?
- 596. What is AppDynamics?

- 597. How do you monitor microservices?
 - 598. What is distributed monitoring?
 - 599. What is centralized logging?
 - 600. What is log aggregation?
-

11. Spring Testing

Testing Fundamentals

- 601. What is unit testing?
- 602. What is integration testing?
- 603. What is the difference between unit testing and integration testing?
- 604. What is `@SpringBootTest` annotation?
- 605. What is `@WebMvcTest` annotation?
- 606. What is `@DataJpaTest` annotation?
- 607. What is `@RestClientTest` annotation?
- 608. What is the difference between `@SpringBootTest` and `@WebMvcTest`?
- 609. When should you use `@SpringBootTest`?
- 610. When should you use `@WebMvcTest`?

Testing with JUnit

- 611. What is JUnit?
- 612. What is JUnit 5?
- 613. What is `@Test` annotation?
- 614. What is `@BeforeEach` and `@AfterEach`?
- 615. What is `@BeforeAll` and `@AfterAll`?
- 616. What is `@DisplayName`?
- 617. What is `@Disabled`?
- 618. What is `@ParameterizedTest`?
- 619. What are assertions in JUnit?
- 620. What is `assertEquals` vs `assertThat`?

Mocking

- 621. What is mocking in testing?
- 622. What is Mockito?
- 623. What is `@Mock` annotation?
- 624. What is `@InjectMocks` annotation?
- 625. What is `@MockBean` annotation?
- 626. What is the difference between `@Mock` and `@MockBean`?
- 627. What is `when().thenReturn()` in Mockito?
- 628. What is `verify()` in Mockito?
- 629. How do you mock void methods?
- 630. What is `ArgumentCaptor`?

Testing REST APIs

- 631. How do you test REST controllers?
- 632. What is MockMvc?
- 633. How do you use MockMvc to test controllers?
- 634. What is @AutoConfigureMockMvc?
- 635. How do you test request and response bodies?
- 636. How do you test HTTP status codes?
- 637. What is TestRestTemplate?
- 638. What is the difference between MockMvc and TestRestTemplate?
- 639. How do you test Spring Security?

Database Testing

- 640. How do you test JPA repositories?
- 641. What is @DataJpaTest?
- 642. What is TestEntityManager?
- 643. How do you use in-memory databases for testing?
- 644. What is H2 database?
- 645. How do you test transactions?
- 646. What is @Transactional in tests?
- 647. What is @Rollback annotation?

Integration Testing

- 648. How do you perform integration testing in Spring Boot?
- 649. What is @SpringBootTest?
- 650. What is @DirtiesContext?
- 651. How do you test with test containers?
- 652. What is Testcontainers library?
- 653. How do you test microservices integration?

12. Spring AOP (Aspect-Oriented Programming)

AOP Fundamentals

- 654. What is AOP?
- 655. What problems does AOP solve?
- 656. What are cross-cutting concerns?
- 657. What are examples of cross-cutting concerns?
- 658. What is the difference between OOP and AOP?
- 659. What are the main concepts in AOP?

AOP Terminology

- 660. What is an aspect?

- 661. What is a join point?
- 662. What is a pointcut?
- 663. What is advice?
- 664. What are the types of advice?
- 665. What is before advice?
- 666. What is after advice?
- 667. What is after returning advice?
- 668. What is after throwing advice?
- 669. What is around advice?
- 670. What is weaving?
- 671. What is the target object?

Spring AOP

- 672. What is @Aspect annotation?
- 673. What is @EnableAspectJAutoProxy?
- 674. How do you create an aspect in Spring?
- 675. What is @Before annotation?
- 676. What is @After annotation?
- 677. What is @AfterReturning annotation?
- 678. What is @AfterThrowing annotation?
- 679. What is @Around annotation?
- 680. What is @Pointcut annotation?
- 681. How do you define pointcut expressions?
- 682. What is execution pointcut?
- 683. What is within pointcut?
- 684. What is @annotation pointcut?
- 685. What is JoinPoint parameter?
- 686. What is ProceedingJoinPoint?

AOP Use Cases

- 687. How do you implement logging using AOP?
- 688. How do you implement security using AOP?
- 689. How do you implement transaction management using AOP?
- 690. How do you implement performance monitoring using AOP?
- 691. What is the difference between Spring AOP and AspectJ?
- 692. When should you use Spring AOP vs AspectJ?

13. Spring Transactions

Transaction Fundamentals

- 693. What is a transaction?
- 694. What are ACID properties?

- 695. What is Atomicity?
- 696. What is Consistency?
- 697. What is Isolation?
- 698. What is Durability?
- 699. What is transaction management?
- 700. What are the types of transaction management in Spring?

@Transactional Annotation

- 701. What is @Transactional annotation?
- 702. What is @EnableTransactionManagement?
- 703. Where can you use @Transactional annotation?
- 704. What are the attributes of @Transactional?
- 705. What is propagation in transactions?
- 706. What are the different propagation types?
- 707. What is REQUIRED propagation?
- 708. What is REQUIRES_NEW propagation?
- 709. What is NESTED propagation?
- 710. What is SUPPORTS propagation?
- 711. What is MANDATORY propagation?
- 712. What is NEVER propagation?
- 713. What is NOT_SUPPORTED propagation?

Transaction Isolation

- 714. What is transaction isolation level?
- 715. What are the different isolation levels?
- 716. What is READ_UNCOMMITTED?
- 717. What is READ_COMMITTED?
- 718. What is REPEATABLE_READ?
- 719. What is SERIALIZABLE?
- 720. What is dirty read?
- 721. What is non-repeatable read?
- 722. What is phantom read?

Transaction Management

- 723. What is rollback in transactions?
- 724. What is rollbackFor attribute?
- 725. What is noRollbackFor attribute?
- 726. What is readOnly transaction?
- 727. What is timeout in transactions?
- 728. What is TransactionTemplate?
- 729. What is PlatformTransactionManager?
- 730. What is declarative transaction management?
- 731. What is programmatic transaction management?
- 732. Which is better: declarative or programmatic transaction management?

14. Microservices Communication

Synchronous Communication

- 733. What is synchronous communication in microservices?
- 734. What are the advantages of synchronous communication?
- 735. What are the disadvantages of synchronous communication?
- 736. How do you implement REST-based communication?
- 737. How do you handle timeouts in synchronous calls?
- 738. How do you implement retry logic?
- 739. What is idempotency in API calls?

Asynchronous Communication

- 740. What is asynchronous communication in microservices?
- 741. What are the advantages of asynchronous communication?
- 742. What are the disadvantages of asynchronous communication?
- 743. What is message-driven architecture?
- 744. What is Apache Kafka?
- 745. What is RabbitMQ?
- 746. What is the difference between Kafka and RabbitMQ?
- 747. When should you use Kafka vs RabbitMQ?

Kafka

- 748. What is Apache Kafka?
- 749. What is a Kafka topic?
- 750. What is a Kafka partition?
- 751. What is a Kafka producer?
- 752. What is a Kafka consumer?
- 753. What is a consumer group?
- 754. What is offset in Kafka?
- 755. What is Kafka broker?
- 756. What is Zookeeper in Kafka?
- 757. How do you integrate Spring Boot with Kafka?
- 758. What is @KafkaListener?
- 759. What is KafkaTemplate?

RabbitMQ

- 760. What is RabbitMQ?
- 761. What is a RabbitMQ exchange?
- 762. What are the types of exchanges?
- 763. What is a direct exchange?
- 764. What is a topic exchange?

- 765. What is a fanout exchange?
- 766. What is a RabbitMQ queue?
- 767. What is binding in RabbitMQ?
- 768. How do you integrate Spring Boot with RabbitMQ?
- 769. What is @RabbitListener?
- 770. What is RabbitTemplate?

Event-Driven Architecture

- 771. What is event-driven architecture?
 - 772. What is an event?
 - 773. What is event sourcing?
 - 774. What is the difference between events and messages?
 - 775. What is publish-subscribe pattern?
 - 776. What is event streaming?
 - 777. What is event store?
 - 778. How do you implement event-driven microservices?
-

15. Microservices Deployment & DevOps

Containerization

- 779. What is containerization?
- 780. What is Docker?
- 781. What is a Docker image?
- 782. What is a Docker container?
- 783. What is Dockerfile?
- 784. How do you create a Docker image for Spring Boot application?
- 785. What is docker-compose?
- 786. What is the difference between Docker and Virtual Machine?
- 787. What are the benefits of using Docker for microservices?

Kubernetes

- 788. What is Kubernetes?
- 789. What is a Kubernetes cluster?
- 790. What is a Kubernetes pod?
- 791. What is a Kubernetes service?
- 792. What is a Kubernetes deployment?
- 793. What is a ReplicaSet?
- 794. What is a ConfigMap?
- 795. What is a Secret in Kubernetes?
- 796. What is Ingress in Kubernetes?
- 797. What is namespace in Kubernetes?
- 798. How do you deploy Spring Boot microservices to Kubernetes?

- 799. What is kubectl?
- 800. What is Helm?

CI/CD

- 801. What is CI/CD?
- 802. What is Continuous Integration?
- 803. What is Continuous Delivery?
- 804. What is Continuous Deployment?
- 805. What is the difference between Continuous Delivery and Continuous Deployment?
- 806. What is Jenkins?
- 807. What is GitLab CI?
- 808. What is GitHub Actions?
- 809. How do you implement CI/CD for microservices?
- 810. What is a build pipeline?

Cloud Platforms

- 811. What is cloud computing?
- 812. What are the types of cloud services?
- 813. What is IaaS, PaaS, and SaaS?
- 814. What is AWS?
- 815. What is Azure?
- 816. What is Google Cloud Platform (GCP)?
- 817. What is AWS Lambda?
- 818. What is serverless architecture?
- 819. How do you deploy Spring Boot to AWS?
- 820. What is Elastic Beanstalk?
- 821. What is ECS (Elastic Container Service)?
- 822. What is EKS (Elastic Kubernetes Service)?

Monitoring & Logging

- 823. What is logging in microservices?
- 824. What is the importance of correlation ID?
- 825. What is centralized logging?
- 826. What is ELK stack?
- 827. What is Elasticsearch?
- 828. What is Logstash?
- 829. What is Kibana?
- 830. What is Splunk?
- 831. How do you implement distributed logging?
- 832. What is APM in microservices?
- 833. What is Prometheus?
- 834. What is Grafana?
- 835. How do you monitor microservices health?

16. Advanced Spring & Microservices Topics

Reactive Programming

- 836. What is reactive programming?
- 837. What is Spring WebFlux?
- 838. What is the difference between Spring MVC and Spring WebFlux?
- 839. When should you use WebFlux?
- 840. What is Mono?
- 841. What is Flux?
- 842. What is the difference between Mono and Flux?
- 843. What is backpressure?
- 844. What is Project Reactor?
- 845. What is reactive streams?

Caching

- 846. What is caching?
- 847. Why is caching important in microservices?
- 848. What is Spring Cache abstraction?
- 849. What is @EnableCaching?
- 850. What is @Cacheable annotation?
- 851. What is @CachePut annotation?
- 852. What is @CacheEvict annotation?
- 853. What is @Caching annotation?
- 854. What are cache providers in Spring?
- 855. What is Redis?
- 856. What is Hazelcast?
- 857. What is Ehcache?
- 858. How do you integrate Redis with Spring Boot?
- 859. What is cache-aside pattern?
- 860. What is write-through cache?
- 861. What is write-behind cache?

Performance & Optimization

- 862. How do you optimize Spring Boot application startup time?
- 863. What is lazy initialization?
- 864. How do you reduce memory footprint?
- 865. What is connection pooling?
- 866. What is HikariCP?
- 867. How do you optimize database queries?
- 868. What is N+1 query problem in JPA?
- 869. How do you use caching to improve performance?
- 870. What is asynchronous processing?

- 871. What is @Async annotation?
- 872. What is @EnableAsync?
- 873. How do you configure thread pools?

Batch Processing

- 874. What is Spring Batch?
- 875. What are the components of Spring Batch?
- 876. What is a Job in Spring Batch?
- 877. What is a Step in Spring Batch?
- 878. What is ItemReader?
- 879. What is ItemProcessor?
- 880. What is ItemWriter?
- 881. What is chunk-oriented processing?
- 882. What is tasklet in Spring Batch?

Scheduling

- 883. What is task scheduling?
- 884. What is @Scheduled annotation?
- 885. What is @EnableScheduling?
- 886. What is cron expression?
- 887. What is fixedRate vs fixedDelay?
- 888. How do you schedule tasks in distributed systems?
- 889. What is Quartz Scheduler?

API Documentation

- 890. What is API documentation?
- 891. What is Swagger?
- 892. What is OpenAPI Specification?
- 893. What is Springdoc?
- 894. What is springdoc-openapi-ui?
- 895. How do you generate API documentation automatically?
- 896. What is @Operation annotation?
- 897. What is @ApiResponse annotation?
- 898. What is @Schema annotation?

Database Migration

- 899. What is database migration?
- 900. What is Flyway?
- 901. What is Liquibase?
- 902. What is the difference between Flyway and Liquibase?
- 903. How do you integrate Flyway with Spring Boot?
- 904. What is version control for databases?
- 905. What is a migration script?

GraphQL

- 906. What is GraphQL?
- 907. What is the difference between REST and GraphQL?
- 908. When should you use GraphQL?
- 909. What is GraphQL schema?
- 910. What is GraphQL query?
- 911. What is GraphQL mutation?
- 912. How do you implement GraphQL in Spring Boot?
- 913. What is Spring GraphQL?

Message Queues Advanced

- 914. What is message acknowledgment?
- 915. What is message durability?
- 916. What is dead letter queue?
- 917. What is message ordering?
- 918. What is exactly-once delivery?
- 919. What is at-least-once delivery?
- 920. What is at-most-once delivery?

Microservices Governance

- 921. What is API versioning?
- 922. What are API contracts?
- 923. What is schema registry?
- 924. What is service mesh governance?
- 925. What is API lifecycle management?
- 926. What is deprecation strategy?

Security Advanced

- 927. What is OAuth 2.1?
- 928. What is PKCE (Proof Key for Code Exchange)?
- 929. What is OpenID Connect?
- 930. What is SAML?
- 931. What is the difference between OAuth and SAML?
- 932. What is multi-factor authentication (MFA)?
- 933. What is single sign-on (SSO)?
- 934. How do you implement SSO in microservices?
- 935. What is zero trust security?
- 936. What is mutual TLS?
- 937. What is API throttling?

Microservices Anti-Patterns

- 938. What are common microservices anti-patterns?

- 939. What is the distributed monolith anti-pattern?
- 940. What is the mega-service anti-pattern?
- 941. What is tight coupling in microservices?
- 942. What is chatty communication?
- 943. What is data coupling?
- 944. What is the shared database anti-pattern?
- 945. What is insufficient monitoring?

DevOps & SRE

- 946. What is DevOps culture?
- 947. What is SRE (Site Reliability Engineering)?
- 948. What is infrastructure as code?
- 949. What is Terraform?
- 950. What is Ansible?
- 951. What is GitOps?
- 952. What is ArgoCD?
- 953. What is observability?
- 954. What are the three pillars of observability?
- 955. What is chaos engineering?
- 956. What is Netflix Chaos Monkey?

Multi-Tenancy

- 957. What is multi-tenancy?
- 958. What are multi-tenancy strategies?
- 959. What is database per tenant?
- 960. What is schema per tenant?
- 961. What is shared schema multi-tenancy?
- 962. How do you implement multi-tenancy in Spring Boot?

Miscellaneous Advanced Topics

- 963. What is feature toggle/feature flag?
- 964. How do you implement feature flags?
- 965. What is A/B testing?
- 966. What is canary release?
- 967. What is backward compatibility?
- 968. What is API deprecation?
- 969. What is polyglot microservices?
- 970. What is serverless microservices?
- 971. What is API rate limiting?
- 972. What is API throttling?
- 973. What is idempotent consumer pattern?
- 974. What is competing consumers pattern?
- 975. What is claim check pattern?
- 976. What is request-reply pattern?

- 977. What is priority queue pattern?
- 978. What is publisher-subscriber pattern?
- 979. What is sequential convoy pattern?
- 980. What is valet key pattern?

Best Practices

- 981. What are Spring Boot best practices?
- 982. What are microservices best practices?
- 983. What are REST API design best practices?
- 984. What are database design best practices for microservices?
- 985. What are security best practices for microservices?
- 986. What are logging best practices?
- 987. What are error handling best practices?
- 988. What are testing best practices?
- 989. What are deployment best practices?
- 990. What are monitoring best practices?
- 991. What is the 12-factor app methodology?
- 992. What are the principles of cloud-native applications?
- 993. What is domain-driven design (DDD)?
- 994. What are bounded contexts in DDD?
- 995. What is the single responsibility principle?
- 996. What is loose coupling and high cohesion?
- 997. What is fail-fast principle?
- 998. What is defensive programming?
- 999. What is the importance of documentation?
- 1000. What are code review best practices?

End of Questions

Note: This comprehensive list contains 1000+ most frequently asked and important Spring Framework, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Spring Cloud, Microservices Architecture, and Design Patterns interview questions covering all major topics.

Tips for Interview Success:

1. **Understand the Concepts Deeply:** Don't just memorize answers; understand the underlying concepts and principles
2. **Practice Hands-On:** Build real projects using Spring Boot and microservices
3. **Stay Updated:** Keep yourself updated with the latest Spring and microservices trends
4. **Real-World Examples:** Be ready with real-world examples from your projects

5. **System Design:** Prepare for system design questions related to microservices architecture
6. **Code Reviews:** Be prepared to write code or explain architectural decisions
7. **Best Practices:** Understand and follow industry best practices
8. **Communication:** Practice explaining complex concepts in simple terms

Topics to Focus Based on Experience Level:

For Freshers (0-2 years): - Spring Core concepts (IoC, DI, Beans) - Spring Boot basics - Spring MVC and REST APIs - Spring Data JPA fundamentals - Basic Spring Security

For Mid-Level (2-5 years): - Advanced Spring Boot features - Microservices fundamentals - Spring Cloud components - Design patterns - Distributed systems concepts - Testing strategies

For Senior/Lead (5+ years): - Microservices architecture and design patterns - System design and scalability - Performance optimization - Security best practices - DevOps and deployment strategies - Team leadership and mentoring

Good luck with your interviews!